

① Continuous & Discrete

Continuous ^{signal variation on} has time ~~variable~~ axis is continuous. whereas discrete signal ~~is~~ is discontinuous.

② Analogy & Digital - Analog can take any value on x axis. Digital has finite values on x axis.

③ Even & Odd - $x_{\text{even}} = \frac{x(t) + x(-t)}{2}$ } $x_{\text{odd}} = \frac{x(t) - x(-t)}{2}$
 $\downarrow$
 $x(t) = x(-t)$ } $x(t) = -x(-t)$

④ Conjugate symmetric and conjugate antisymmetric
 $\downarrow$ $\downarrow$
 $x(t) = x^*(t)$ } $x(t) = -x^*(-t)$

⑤ Periodic & Non Periodic
 $\downarrow$
 $x(t) = x(t + T)$

Q. Fundamental time period of $x(t) = A_0 e^{j\omega t}$
 $x(t+T) = A_0 e^{j\omega(t+T)}$ $A_0 (\cos j\omega t + j \sin j\omega t)$
 $j\omega(t+T) = j\omega t$
 $T = \frac{2\pi}{\omega_0}$

If $x(t) = x_1(t) + x_2(t)$
 $\downarrow$ $\downarrow$
 (T_1, ω_1, f_1) (T_2, ω_2, f_2)

$\frac{T_1}{T_2} \propto \frac{\omega_1}{\omega_2} \propto \frac{f_1}{f_2} = \text{rational no.}$

Then, $T = \text{LCM}[T_1, T_2]$
 $\omega_0 = \text{HCF}[\omega_1, \omega_2]$
 $f_0 = \text{HCF}[f_1, f_2]$

Q. find period of $x(t) = \sin 2\pi t + \cos 8\pi t$
 ~~$T = 2\pi$~~ , $\omega_0 = \text{HCF}(2\pi, 8\pi) = 2\pi$
 $\Rightarrow T = 1$

Q. find fundamental time period of $\sin 4\pi t + \cos 2t$?
 $\rightarrow$ Non periodic as $\frac{\omega_1}{\omega_2} = \frac{2\pi}{\pi}$ which is irrational

Energy and power signals-
 for any signal $x(t)$.

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt.$$

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt. \quad \rightarrow \text{for periodic.}$$

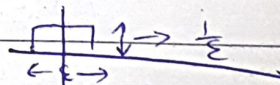
$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_T |x(t)|^2 dt \quad \rightarrow \text{for non periodic.}$$

$\rightarrow$ for Energy signal $P=0$ & E is finite.
 $\Rightarrow$ Energy signals are integrable (Energy is finite)

$\rightarrow$ for power signal, power is finite, Energy $= \infty$
 periodic power signals should be absolutely integrable over their time period.

BASIC SIGNALS-

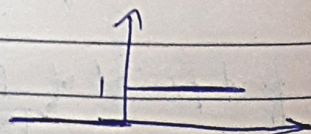
① Unit step function-
 area of this is always 1.



$$\delta(a(t-t_0)) = \frac{1}{|a|} \delta(t-t_0)$$

② Unit step signal-

$$x(t) = u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

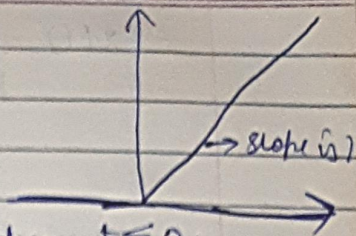


④ Unit Ramped signal-

$$x(t) = t u(t)$$

$$x(t) = \int_{-\infty}^t u(t) dt$$

$u(t)$ is 0 is when $t < 0$.



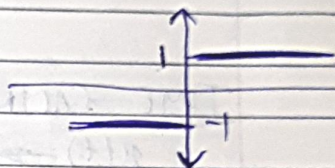
⑤ Ramped Parabolic function-

$$p(t) = \frac{t^2}{2} u(t)$$

$$= \int_{-\infty}^t x(t) dt$$

⑥ Signum function-

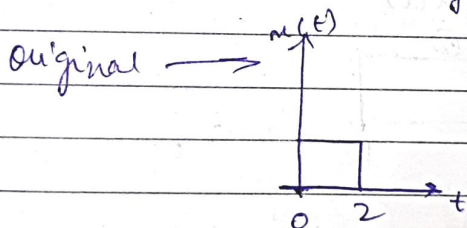
$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$$



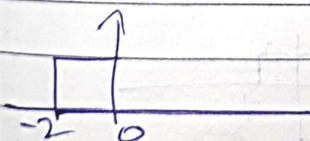
OPERATIONS ON SIGNAL-

① SHIFTING- ① in time ② in amplitude.

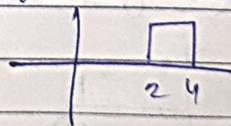
② TIME SHIFTING- $x(t) = y(t+k)$; $k > 0$ (left shift); $k < 0$ (right shift)



if left shift



if right shift

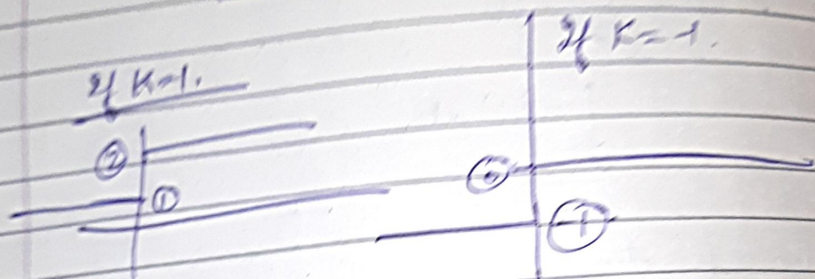
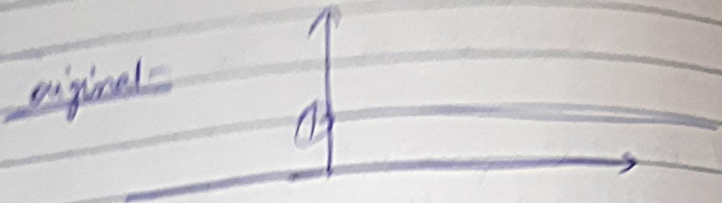


③ Amplitude shifting

$$x(t) = y(t) = Kx(t)$$

$$x(t) = y(t) + K y(t)$$

$K > 0$ upward shift
 $K < 0$ downward shift

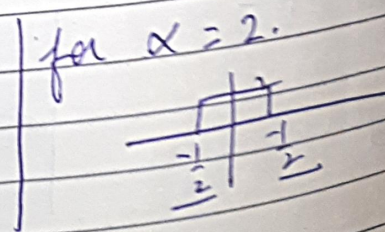
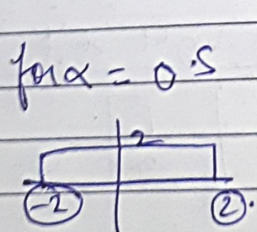
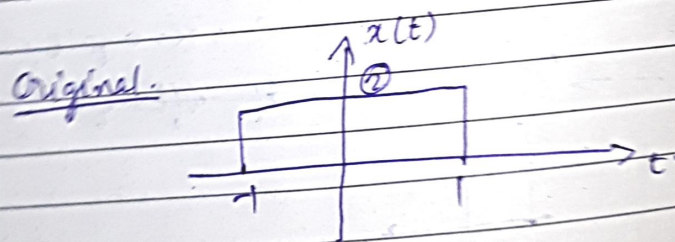


TIME SCALING

$$x(t) \rightarrow y(t) = x(\alpha t) \quad \text{when } \alpha \neq 0$$

$0 < \alpha < 1 \rightarrow$ time expansion

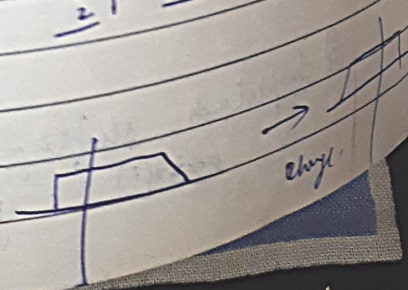
$\alpha > 1 \rightarrow$ time compression



TIME REVERSAL

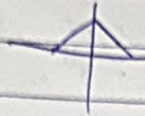
$$y(t) = x(-t)$$

original

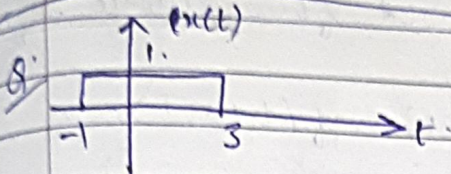
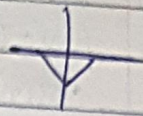


AMPLITUDE REVERSAL

Original.

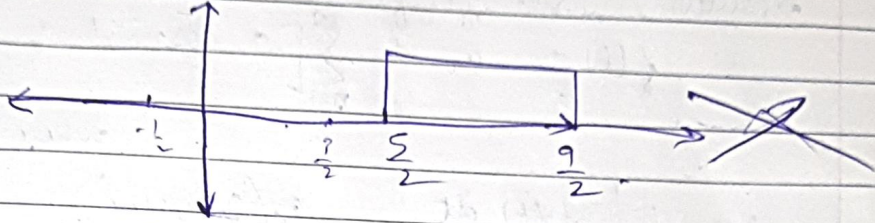


changed.

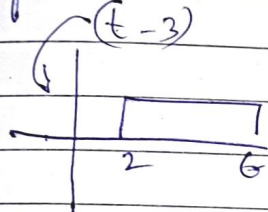


draw $x(2t-3)$.

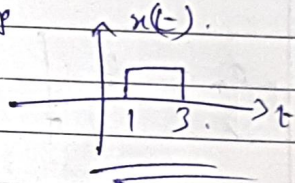
Ans



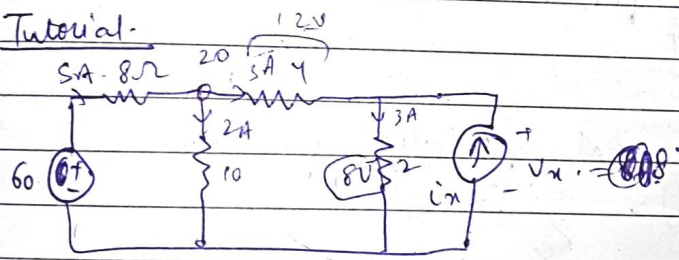
→ first do time shift.



Then do scaling



Tutorial.

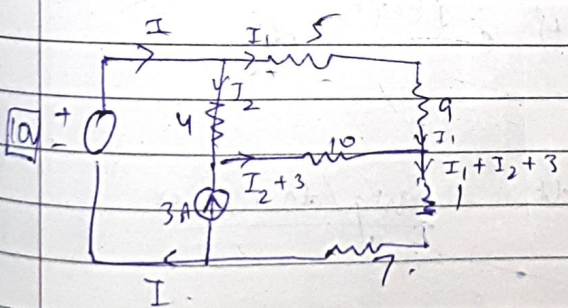


$$\begin{cases} i_x = 1 \\ V_x = 8 \end{cases}$$

$$\frac{10 \times 6}{16} = \frac{5}{4}$$

$$8 + \frac{15}{4}$$

$$\frac{47}{4}$$



$$5I_1 + 9I_1 - 10(I_2 + 3) - 4I_2 = 0$$

$$10 \frac{15}{4}$$

$$\frac{15}{4}$$

$$-10 + 5I_1 + 9I_1 + (I+3) + 7(I+3) = 0$$

Fourier Series & Fourier Transform

- (1) Fourier series is an accurate representation of a periodic signal and consist of sum of sinusoids and cosinusoids at fundamental and harmonic frequencies.
- (2) Fourier series expansion is used for periodic signals.

Consider periodic function $f(t) = f(t + nT)$.

→ Fourier series for $f(t)$

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$$

where $\omega_0 = \frac{2\pi}{T}$

$$a_0 = \frac{1}{T} \int_T f(t) dt \quad a_0 (\text{DC value})$$

$$a_n = \frac{2}{T} \int_T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_T f(t) \sin(n\omega_0 t) dt$$

→ If $f(t)$ is real periodic function.

$$f(t) = \sum_{-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$\Rightarrow C_n = \frac{1}{T} \int_T f(t) e^{-jn\omega_0 t} dt$$

$$C_0 = \frac{1}{T} \int_T f(t) dt = \underline{\text{avg/dc value.}}$$

Conditions for existence of Fourier series expansion:

- ① Signal should be deterministic over its time period.

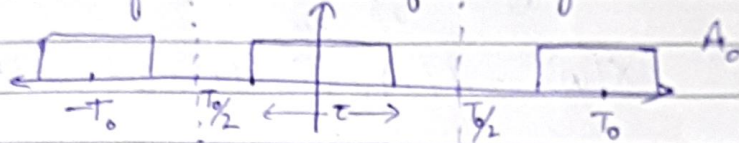
it should have finite number of maximas and minimas.

② Signal should have finite of discontinuities over its time period.

③ A signal must be integrable over its time period.

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty$$

5. Determine fourier series of this signal.



we will use exponential function as A_0 is purely real.

$$C_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-j n \omega_0 t} dt$$

$$\omega_0 = \frac{2\pi}{T}$$

$$= \frac{A_0}{T_0} \int_{-\tau/2}^{\tau/2} e^{-j n \omega_0 t} dt$$

$$= \frac{A_0}{T_0} \left[\frac{e^{-j n \omega_0 t}}{-j n \omega_0} \right]_{-\tau/2}^{\tau/2}$$

$$T = T_0$$

$$= \frac{A_0}{T_0} \left[\frac{e^{-j n \omega_0 \tau/2} - e^{j n \omega_0 \tau/2}}{-j n \omega_0} \right]$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$= \frac{2A_0}{T_0} \times \left[\frac{1}{n \omega_0} \sin\left(\frac{n \omega_0 \tau}{2}\right) \right]$$

$$\text{Put } \omega_0 = \frac{2\pi}{T_0}$$

$$C_n = \frac{A_0 \tau}{n \pi} \sin\left(\frac{n \pi \tau}{T_0}\right)$$

$$C_0 = \frac{A_0 \tau}{T_0} = \frac{A_0}{2} ; C_1 = \frac{A_0}{\pi}$$

FOURIER TRANSFORM.

It is used for frequency analysis of a signal. For periodic signal, we have representation of signal as linear combination of complex exponentials. For ~~aperiodic~~ nonperiodic signal, we have same representation in terms of Fourier transform.

$$x(t) = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$X(\omega) = F(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Same conditions hold for Fourier series as sufficient conditions for convergence of Fourier transform.

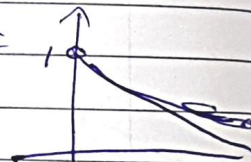
Q. $x(t) = e^{-at} u(t)$; $a > 0$.
Plot magnitude and phase plot.

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-at-j\omega t} dt$$

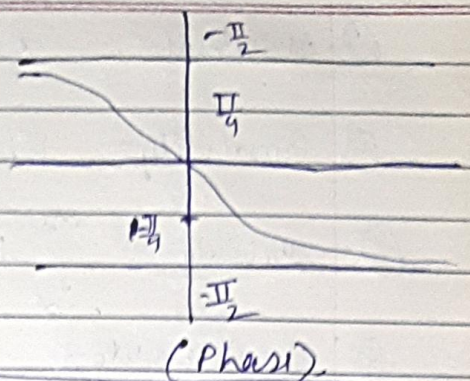
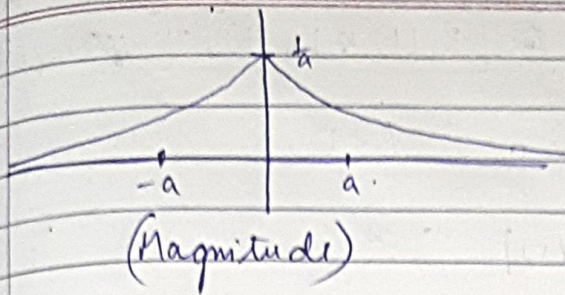
$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{1}{j\omega + a}$$



$$|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$

$$\angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$



LTI system and its properties- linear time invariant system.

The system is linear if it holds principle of superposition & proportionality. The time invariant means any delay provided in the input must reflect in the output.

Invariant $\rightarrow$ does not depend on ~~time~~ disturbance ~~existence~~

Convolution is a mathematical tool

$$y(t) = x(t) \otimes h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$y(n) = x(n) \otimes h(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} h(k) x(n-k)$$

Transfer function is defined as ratio of Laplace transform of the output to input when all initial conditions are zero.

$$H(s) = \frac{Y(s)}{X(s)}$$

Total response = $Z_{IR} + Z_{SR}$
zero input response $\rightarrow$ zero state response.

PROPERTIES OF SYSTEM-

① Commutative property. $x(t) \otimes h(t) = h(t) \otimes x(t)$

② Distributive $\rightarrow x(t) \otimes [h_1(t) + h_2(t)] = x(t) \otimes h_1(t) + x(t) \otimes h_2(t)$

③ Associative - $x(t) \otimes [h_1(t) \otimes h_2(t)] = [x(t) \otimes h_1(t)] \otimes h_2(t)$

④ Causality - $h(t) = 0, t < 0$.

⑤ Stability - $\int_{-\infty}^{\infty} |h(t)| dt < \infty$.

⑥ Invertibility -

$$x(t) \xrightarrow{[h(t)]} [h_y(t)] \rightarrow y(t) = x(t).$$

$$h(t) \otimes h_y(t) = \delta(t).$$

$$\rightarrow h(s) = \frac{1}{h_y(s)}.$$

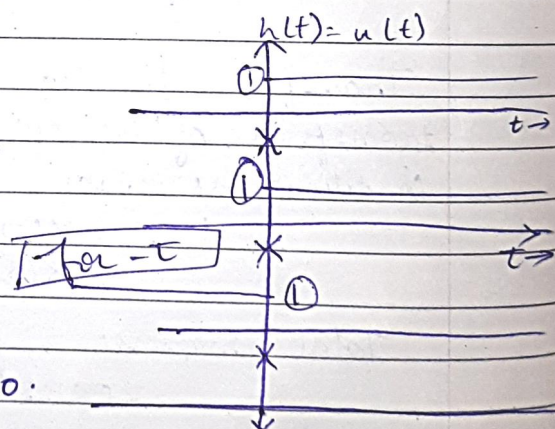
Q. $x(t) \xrightarrow{[h(t)]} y(t)$

$$y(t) = x(t) \otimes h(t)$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$\rightarrow x(t) = u(t); h(t) = u(t); y(t).$$

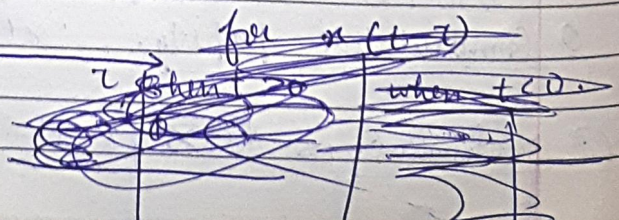
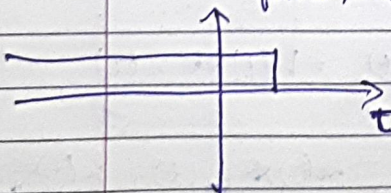
$$\Rightarrow y(t) = \int_{-\infty}^{\infty} u(\tau) u(t-\tau) d\tau.$$



for $x(t-\tau)$.

When $t > 0$.
for example $x(2-\tau)$

When $t < 0$.



RAM P

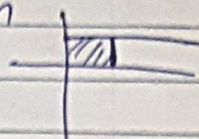
for $x(t-T)$ when $t > 0$

0 because both are 0 in each other's area

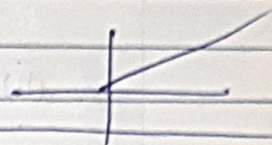
for $x(t-T)$ when $t < 0$

area in common

$$\int 1 \cdot 1 dt = T$$

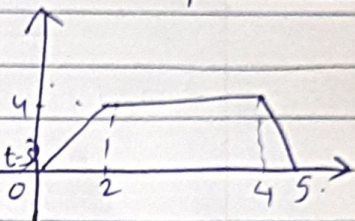


$$y(t) = t$$



Q. find $x(t)$?

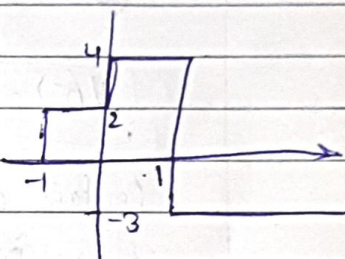
$$x(t) = 2x(t) - 2x(t-2) - 4x(t-4) + 4x(t-6)$$



Q. find $x(t)$.

$$x(t) = 2\delta(t+1) + 4\delta(t) - 3\delta(t-1)$$

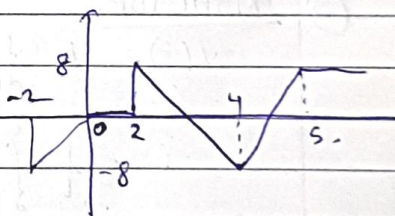
$$2u(t+1) + 2u(t) - 7u(t-1)$$



Q. find $x(t)$?

$$-8u(t+2)$$

$$x(t) = +4x(t+2) - 4x(t) + 8u(t+2) - 8x(t-2) - 12u(t-4) - 18x(t-5)$$



① RESISTOR

$$① V(t) = Ri(t)$$

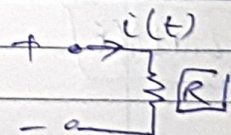
$$② \text{ in Laplace domain } V(s) = I(s) \cdot R$$

$$③ \text{ in phasor domain } V = |V| \angle \theta$$

$$I = |I| \angle \phi$$

$$\rightarrow V = IR$$

$\phi = 0$ for resistor



~~2.11~~

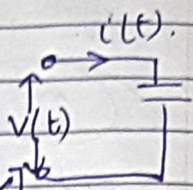
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② CAPACITOR -

$$(i). i(t) = \frac{1}{C} \frac{dV(t)}{dt}$$

$$V(t) = \frac{1}{C} \int_0^t i(t) dt + \underbrace{V(0^-)}_{I.C.} \quad [TIME]$$



$$(ii). V(s) = \frac{1}{sC} (I s)$$

$$Z_C = \frac{1}{sC}$$

(LAPLACE).

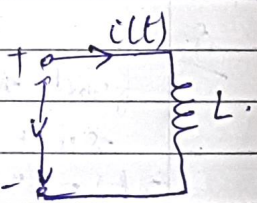
$$(iii). \text{ for phasor, } CV = \left(\frac{1}{j\omega C}\right) I$$

$V(0^-) = V(0^+)$ $\therefore$ Voltage across capacitor does not change instantaneously if $i(t)$ doesn't contain any impulse or derivative of impulse function.

③ INDUCTOR -

$$V(t) = L \frac{di(t)}{dt}$$

$$i = \frac{1}{L} \int_0^t V(t) dt + \underbrace{i(0^-)}_{I.C.} \quad [TIME]$$



LAPLACE

$$V(s) = sL I(s)$$

$$Z_L = sL$$

PHASOR

$$V = j\omega L I$$

Voltage must not contain impulse or derivative of impulse.

TRANSIENT ANALYSIS of first order circuit.

- ① Circuit that contain capacitor & inductor can be represented by differential equation whose order is usually equal to the number of capacitor + inductor in the circuit. @
- ② Circuits that contain only 1 energy storing element can be represented by 1st order differential eqⁿ. Hence they are called first order circuit.
- ③ In general, the complete response of first order circuit consists of sum of 2 parts, the natural and forced response.
- ④ The complete response of 1st order circuit will ~~also~~ depend on initial condition, usually a capacitor voltage or inductor current in a particular time. Let t_0 denote the time at which initial condition is given, Then the natural response of first order circuit will be of the following form

$$NR = Ke^{-\frac{(t-t_0)}{\tau}}$$

when $t_0 = 0$; $NR = Ke^{-\frac{t}{\tau}}$

where constant K depends upon initial conditions.

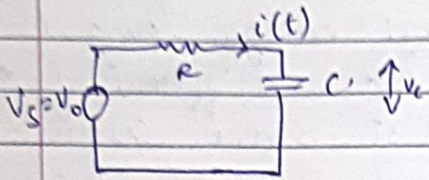
- ⑤ In the case when input is a constant or sinusoid, the forced response is also called as steady state response and natural response is called as transient response. Hence, complete response is addition of steady state and transient.

STEPS TO GET COMPLETE RESPONSE OF FIRST ORDER:

- ① find the forced response, before the disturbance, evaluate this response at time $(t-t_0)$, to obtain

initial condition of energy storing element.

- ② find the forced response after the disturbance.
- ③ Add the natural response to the forced response, to get complete response and use initial conditions to evaluate the value of constant K .

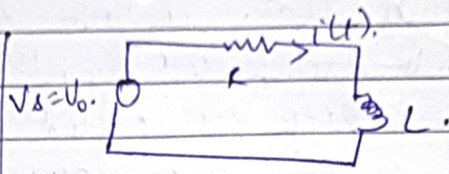


$$-V_0 + Ri(t) + V_c(t) = 0$$

~~$$-V_0 + Ri(t) +$$~~

$$-V_0 + RC \frac{dV_c(t)}{dt} + V_c(t) = 0$$

$$\boxed{\frac{dV_c(t)}{dt} + \frac{V_c(t)}{RC} = \frac{V_0}{RC}}$$



$$-V_0 + Ri(t) + L \frac{di(t)}{dt} = 0$$

$$\frac{L di(t)}{dt} + Ri(t) = V_0$$

$$\Rightarrow \boxed{\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_0}{L}}$$

$$\boxed{\frac{dx(t)}{dt} + \frac{x(t)}{\tau} = K} \rightarrow$$

In Capacitor, V_c is ~~current~~ taken in diff equation.
In Inductor, I_L is taken in diff equation.

$$\frac{dx(t)}{dt} = \frac{K\tau - x(t)}{\tau}$$

$$\frac{dx}{K\tau - x} = \frac{dt}{\tau}$$

$$\Rightarrow \int \frac{dx}{x - K\tau} = -\int \frac{dt}{\tau}$$

$$\Rightarrow \ln(x - K\tau) = -\frac{1}{\tau} \int dt + D$$

$$\Rightarrow x - K\tau = e^{-t/\tau} e^D$$

$$\Rightarrow \boxed{x = A e^{-t/\tau} + K\tau}$$

$A = e^D \rightarrow \text{Initial Condition}$

$$x(0) = K\tau + Ae^0 = K\tau + A$$

$$\Rightarrow A = x(0) - K\tau$$

$$\Rightarrow x(t) = K\tau + [x(0) - K\tau] e^{-t/\tau}$$

Since $\lim_{t \rightarrow \infty} x(t) = K\tau$

$$\Rightarrow x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$

Calculation of Time Constant.

$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$

$$\frac{d}{dt}(x(t)) = -\frac{1}{\tau} [x(0) - x(\infty)] e^{-t/\tau}$$

Let $t=0$.

$$\left. \frac{d}{dt}(x(t)) \right|_{t=0} = -\frac{1}{\tau} [x(0) - x(\infty)]$$

$$\Rightarrow \tau = \frac{x(\infty) - x(0)}{\left. \frac{d}{dt}(x(t)) \right|_{t=0}}$$

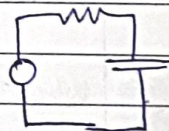
~~Ex~~ In a capacitive circuit

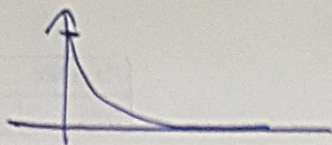
$$\frac{d}{dt} V(t) + \frac{V(t)}{RC} = \frac{V_0}{RC}$$

$$\frac{d}{dt} x(t) + \frac{x(t)}{RC} = \frac{V_0}{RC}$$

$$\tau = RC \quad K = \frac{V_0}{\tau}$$

$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$





$$V(t) = \underbrace{V_0}_{FR.} + \underbrace{[V(0) - V_0]}_{NR.} e^{-t/\tau}$$

for inductor -

$$\frac{d I_L(t)}{dt} + R \frac{I_L(t)}{L} = \frac{R V_0}{L}$$



$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$

$$I_L(t) = R\tau + \left[\frac{V(0)}{R} - I_0 \right] e^{-t/\tau}$$

$$\Rightarrow I_L(t) = \underbrace{I_0}_{\text{Forced.}} + \underbrace{[I(0) - I_0]}_{\text{Natural.}} e^{-t/\tau}$$

$$I_0 = \frac{V_0}{R}$$

STABILITY of first order circuit.

$$x_n = k e^{-t/\tau}$$

$$\text{Total} = x(t) = x_n(t) + x_f(t).$$

→ When τ is +ve, natural response vanishes and only forced response is left. This case is said to be stable condition.

→ When τ is negative, the natural response rises beyond time. $\therefore$ Circuit becomes unstable.

Circuit containing OPAMP and dependent sources may have -ve time constant.

Response of first ^{order} circuit to non constant input

$$\frac{d(x(t))}{dt} + \frac{x(t)}{\tau} = y(t)$$

$$\frac{d(x(t))}{dt} + a(x(t)) = y(t)$$

where $a = \frac{1}{\tau}$

$$y(t) = \text{FR} = e^{-at} \int y(t) e^{-at} dt.$$

$$\text{NR} = k e^{-at}$$

↓
any function

Q. $y(t) = e^{bt}$

$\Rightarrow FR = \frac{e^{bt}}{a+b} \quad | \quad NR = Ke^{-at}$

When forcing function is sinusoidal-

$\frac{d}{dt} x(t) + ax(t) = y(t) \quad \text{--- (2)}$
Let $y(t) = M \sin(\omega t + \phi)$

$x(t) = x_N(t) + x_f(t)$

$x_N(t) = Ke^{-at}$

$x_f(t) = e^{-at} \int M \sin(\omega t + \phi) e^{at} dt$

ONLY FOR SINUSOIDAL $\Rightarrow x_f(t) = \underbrace{A \sin \omega t + B \cos \omega t}_{\text{constants}}$

SUMMARY OF FIRST ORDER CIRCUITS

SNO	$y(t)$	$x_{\text{forced}}(t)$
①	Constant $y(t) = M$	$x_f(t) = N$ where N is constant.
②	$y(t) = Me^{bt}$	$x_f(t) = Ne^{bt}$ N is constant
③	$y(t) = Me^{-bt}$	$x_f(t) = Ne^{-bt}$
④	<u>Sinusoidal</u> $y(t) = M \sin(\omega t + \phi)$	$x_f(t) = A \sin \omega t + B \cos \omega t$
⑤	$y(t) = e^{-bt}$	$x_f(t) = \frac{e^{-bt}}{a-b}$ If $a=b$ Then $x_f(t) = t e^{-bt}$

Calculation of Time Constant for RL Circuit

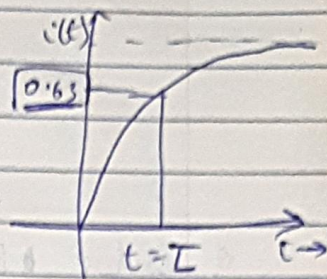
(11)
1st

$$i(t) = \frac{V_0}{R} + \left[i(0) - \frac{V_0}{R} \right] e^{-\frac{t}{\tau}}$$

If $i(0) = 0$.

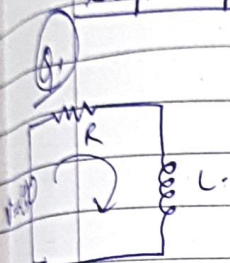
$$i(t) = \frac{V_0}{R} + \left[0 - \frac{V_0}{R} \right] e^{-t/\tau}$$

$$i(t) = \frac{V_0}{R} [1 - e^{-t/\tau}]$$



if initial value is 0

Step response of RL circuit.



Initial current = 0.

$$-V_0 + IR + L \frac{di}{dt} = 0$$

$$\Rightarrow \frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_0}{L}$$

$$\frac{di(t)}{dt} = \frac{V_0}{L} - \frac{R}{L} i(t)$$

$$\frac{di(t)}{V_0 - Ri(t)} = \frac{dt}{L}$$

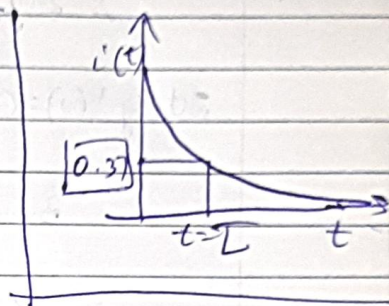
$$\Rightarrow \int \frac{di(t)}{V_0 - Ri(t)} = \int \frac{dt}{L}$$

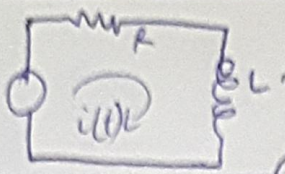
$$I_L = KI + [i(0) - KI] e^{-t/\tau}$$

$$i(0) = 0 \Rightarrow K + \frac{1}{2} \Rightarrow K = -\frac{1}{2}$$

$$= KI [1 - e^{-t/\tau}]$$

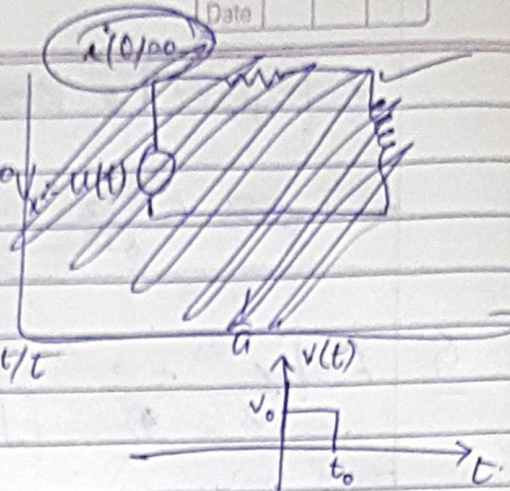
$$= \frac{1}{R} (1 - e^{-t/\tau})$$





$$V(t) = \begin{cases} 0 & t \leq 0 \\ V_0 & 0 < t < t_0 \\ 0 & t \geq t_0 \end{cases}$$

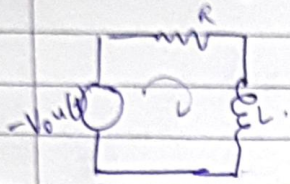
Pulse source.



$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$

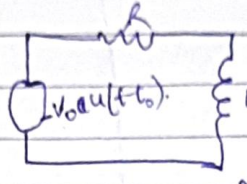
Ans

$V(t) \Rightarrow$ ①



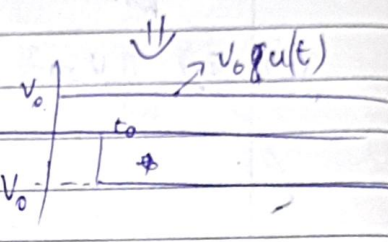
$$i_1(t) = \frac{V_0}{R} [1 - e^{-t/\tau}]$$

②



$$i_2(t) = \frac{V_0}{R} [1 - e^{-(t-t_0)/\tau}]$$

$$I(0) = 0$$



$$i_{total} = i_1 + i_2$$

$$= \frac{V_0}{R} [1 - e^{-t/\tau}] - \frac{V_0}{R} [1 - e^{-(t-t_0)/\tau}]$$

$$0 + [0 - \frac{V_0}{R}] e^{-t/\tau}$$

$$\frac{V_0 - V_0}{R} e^{-t/\tau}$$

$$\frac{V_0}{R} [e^{-(t-t_0)/\tau} - e^{-t/\tau}]$$

$$i_{total} = \frac{V_0}{R} (e^{-t/\tau}) [e^{t_0/\tau} - 1]$$

$$\frac{V_0}{R}$$

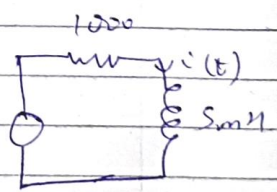
at $t = t_0$,

$$\Rightarrow i_{total} = \frac{V_0}{R} [1 - e^{-t/\tau}]$$

Q. find $i(t) =$

$$i(0) = 0 ; i(\infty) = 4 \times 10^{-3}$$

$$\Rightarrow i_L(t) = 4 \times 10^{-3} + [-4 \times 10^{-3}] e^{-t/5 \times 10^{-6}}$$



$$\frac{5 \times 10^{-3}}{10^{-2}}$$

$$= 4 \times 10^{-3} [1 - e^{-(2 \times 10^5)t}]$$

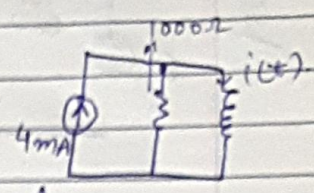
By source transformation

$$i(t) = i(s) + [i(0) - i(s)]e^{-t/\tau}$$

$$i(0) = 4 \times 10^{-3} \text{ A} \quad i(\infty) = 0$$

$$\therefore i(t) = 4 \times 10^{-3} - 4 \times 10^{-3} e^{-t/(5 \times 10^{-4})}$$

$$= 4 \times 10^{-3} [1 - e^{-(2 \times 10^3)t}]$$

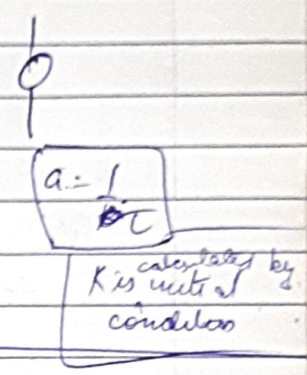


By I_n & I_f Method

$$I_n = Ke^{-t/\tau}$$

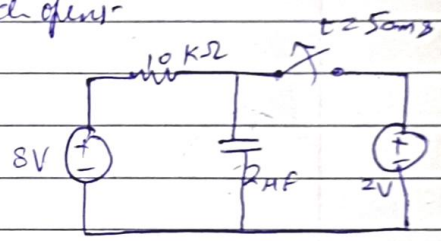
$$I_f = e^{-at} \int y(t) e^{at} dt$$

$$= e^{-at} \int_0^t \frac{y}{L} e^{-at} dt$$



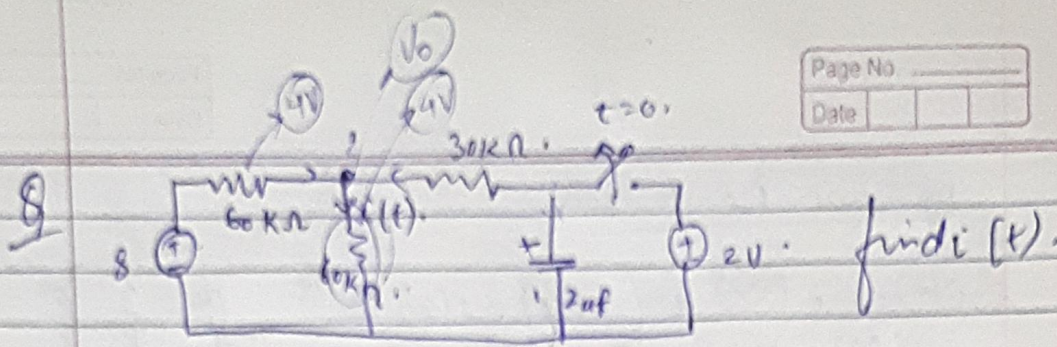
- Q. 1. V_c when switch opens.
2. V_c after 50ms after switch opens.

Ans 1. Now $t_0 = 50 \text{ ms}$.



Before t_0 , Capacitor is acting as open circuit. $\therefore$ Its voltage $V_c = 2V$

at $t = \infty$, $V_c(\infty) = 8V$ $\therefore V(50\text{ms}) = 2V$



$v_L(0^-) = 2V$ $V_0 = 4V$

$v_L(0) = 2$ $V_L(\infty) = 4$

$v(t) = 4 + (2 - 4)e^{-t/0.12}$
 $= 4 - 2e^{-t/0.12}$

$i(t) = \frac{dv}{dt}$

$\frac{V_0 - 8}{60K} + \frac{V_0}{60K} + \frac{V_0 - (4 - 2e^{-t/0.12})}{30K} = 0$

$V_0 - 8 + V_0 + 2V_0 - 8 + 2e^{-t/0.12} = 0$

$4V_0 = 16 - 2e^{-t/0.12}$

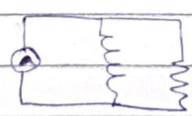
$V_0 = 4 - \frac{1}{2}e^{-t/0.12}$

$\frac{V_0}{60K} = 6.66 \times 10^{-5} + [16.7 \times 10^{-6}]e^{-t/0.12}$

SEQUENTIAL SWITCHING -

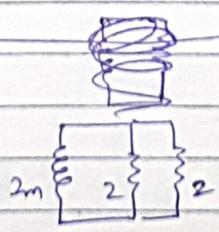
Q. find $i(t)$?
 $t < 0$

$i(0^-) = 10A$

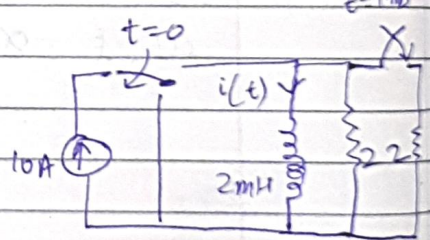


~~$i(t)$~~

$t > 1ms$



$0 < t < 1$
 $i(t) = 10e^{-t}$
 $i(t) \text{ at } t=1 \Rightarrow \frac{10}{e} = 3.68$



$$i(t) = 0 + [3.68 - 0] e^{-(t-1)/2}$$

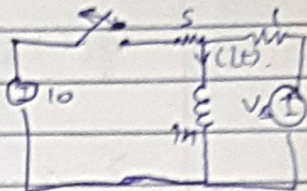
$$\Rightarrow i(t) = 3.68 e^{-(t-1)/2}$$

Q. find current $i(t)$

$$v(t) = 10 e^{-2t} u(t)$$

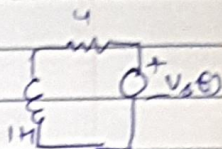
$$t < 0$$

$$i(0^-) = 2$$



$$\text{at } t = 0$$

$$-10 e^{-2t} + 4 i(t) + \frac{di(t)}{dt} = 0$$



$$i(t) = e^{at} \int y(t) e^{-at} dt$$

$$\Rightarrow \frac{di(t)}{dt} + 4 i(t) = 10 e^{-2t}$$

$$\Rightarrow i_f(t) = A e^{-2t}$$

$$\Rightarrow A = 5$$

$$i(t) = i_N(t) + i_f(t)$$

$$\Rightarrow i(t) = 5 e^{-2t} + K e^{-4t}$$

$$i(0) = 2 = 5 + K$$

$$\Rightarrow K = -3$$

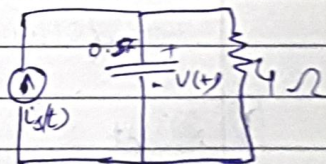
$$i_N = K e^{-4t}$$

$$i_N = K e^{-4t}$$

$$i(t) = 5 e^{-2t} - 3 e^{-4t}$$

Q. find $v(t)$ for $t > 0$ if $i_s = 10 \sin 2t u(t)$, given $v(0) = 0$.

$$i(0) = 0$$

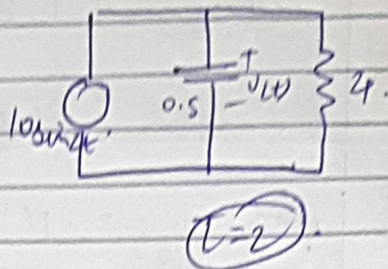


Use I forced and I natural if sinusoidal or exponential function on RHS is given.

$$10 \sin 2t = \frac{V(t)}{4} + \frac{1}{2} \frac{dV(t)}{dt}$$

$$\rightarrow \frac{dV(t)}{dt} + \frac{V(t)}{2} = 20 \sin 2t$$

$$V_f(t) = A \sin 2t + B \cos 2t$$



$$\rightarrow A = \frac{40}{17} ; B = -\frac{160}{17}$$

$$\rightarrow V_{\text{forced}}(t) = \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$

$$V_{\text{natural}}(t) = K e^{-t/2}$$

$$V(t) = K e^{-t/2} + \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$

at $t=0$.

$$V(0) = 0 = K + 0 - \frac{160}{17} \quad \Rightarrow K = \frac{160}{17}$$

$$V(t) = \frac{160}{17} e^{-t/2} + \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$

2nd Order Circuit.

Circuit that contain 2 energy storing elements are characterised by 2nd order differential equation. we use following methods to develop 2nd order equation

- ① Direct ② Operator ③ State Variable.

① DIRECT - steps -

① Identify first and second variable x_1 & x_2 . Then

$$I_L / V_C$$

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Date _____

variable are inductor current ~~and~~ capacitor voltage.

② write 1st order differential equation obtaining $\frac{dx_1}{dt} = f(x_1, x_2)$.

③ Obtain 1 additional differential equation which is of first order but in terms of 2nd order.

Such that $\frac{dx_2}{dt} = Kx_1$

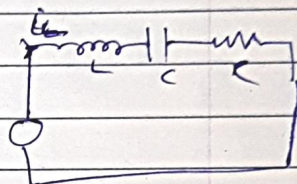
④ Put the values obtained in step 3 to the equation in step 2 and thus obtaining 2nd order differential eq.

Example 1 here $x_1 = I_L$; $x_2 = V_C$.

$$-V_{S5} + L \frac{di(t)}{dt} + V_C + iR = 0.$$

$$\frac{dV_C}{dt} = i = C \frac{dV_C}{dt}$$

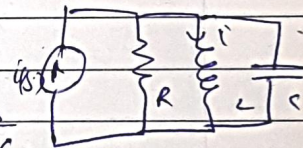
$$\Rightarrow \frac{d^2 V_C(t)}{dt^2} + R \frac{dV_C(t)}{L dt} + \frac{V_C(t)}{LC} = \frac{V_S}{LC}$$



Example 2

$$I_R + I_L + I_C = I_S$$

$$\Rightarrow \frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di(t)}{dt} + \frac{di(t)}{LC} = \frac{I_S}{LC}$$



OPERATOR METHOD -

Step

- ① Identify variable for which soln is to be obtained.
- ② write differential equation in terms of ~~desired~~ desired variable x_1 and 2nd variable x_2 .
- ③ obtain additional equation in terms of 1st and 2nd variables.
- ④ use operator. So that $\frac{1}{s} = \int dt$ to obtain algebraic equation in terms of s and 2 variables x_1 & x_2 .

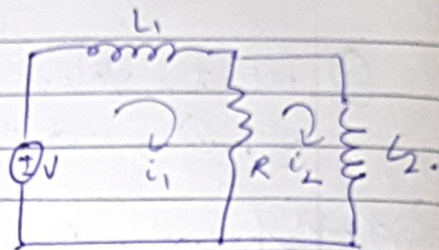
5) Use Cramer's rule, to solve the desired variable, so that $x_1 = f(s, \text{sources}) = \frac{P(s)}{Q(s)}$.

6) Obtain $Q(s) x_1 = P(s)$.

7) Convert back the operator for the equation involved in step 6 and thus obtain 2nd order diff eqⁿ.

Q. $R = 1\Omega$, $L_1 = 1H$, $L_2 = 2H$.

Ans. Let $\frac{d}{dt} = s$.
 $\int dt = \frac{1}{s}$.



$$-V_s + L_1 \frac{di_1}{dt} + R(i_1 - i_2) = 0 \quad \text{--- (1)} \quad \left| \quad L_2 \frac{di_2}{dt} + R(i_2 - i_1) = 0 \quad \text{--- (2)} \right.$$

$$\begin{aligned} \rightarrow -V_s + s i_1 + i_1 - i_2 &= 0 & \left| \quad 2s i_2 + i_2 - i_1 &= 0. \right. \\ \rightarrow (s+1)i_1 - i_2 &= V_s \\ \rightarrow -i_1 + (2s+1)i_2 &= 0. \end{aligned}$$

Apply Cramer's rule and find i_1 & i_2 .

$$\begin{bmatrix} s+1 & -1 \\ -1 & 2s+1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix}.$$

$$\rightarrow i_2 = \frac{V_s}{(s+1)(2s+1)-1}$$

$$\rightarrow V_s = i_2 [2s^2 + 3s]$$

$$\rightarrow V_s = 2 \frac{d^2 i_2}{dt^2} + 3 \frac{di_2}{dt}$$

$$\psi = \frac{1}{L} \int \psi dt$$

SOLUTION of 2nd order diff. eqⁿ.

$$a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = f(t).$$

2/ $f(t) = 0$ [$\therefore$ we are only considering natural response solution]

$$\text{Let } x_n = A e^{st}$$

$$\Rightarrow a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = 0.$$

$$\Rightarrow a_2 \frac{d^2 [A e^{st}]}{dt^2} + a_1 \frac{d[A e^{st}]}{dt} + a_0 [A e^{st}] = 0.$$

$$\Rightarrow a_2 s^2 A e^{st} + a_1 s A e^{st} + a_0 A e^{st} = 0.$$

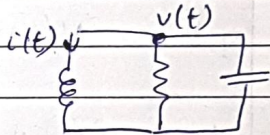
$$\Rightarrow \underbrace{A e^{st}}_{x_n} [a_2 s^2 + a_1 s + a_0] = 0.$$

$$\Rightarrow s_1, s_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}$$

$$\Rightarrow x_n = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Apply KVL-

$$\frac{V}{R} + i(t) + C \frac{dv}{dt} = 0.$$



$$\Rightarrow \frac{V}{R} + \frac{1}{L} \int_0^t V dt + C \frac{dv}{dt} + i(0) = 0$$

initial current.

$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} (V) = 0.$$

$\Rightarrow$ Take Laplace.

$$C [s^2 V(s)] + \frac{1}{R} V(s) + \frac{V(0)}{L} = 0.$$

$$\Rightarrow V(s) \left[s^2 C + \frac{1}{R} + \frac{1}{L} \right] = 0.$$

!!

~~Find the~~ $\rightarrow a_2 s^2 + a_1 s + a_0 = 0$
 h.w, $a_2 = C$, $a_1 = \frac{1}{R}$, $a_0 = \frac{1}{LC}$
 $s_1, s_2 = \frac{-\frac{1}{R} \pm \sqrt{\left(\frac{1}{R}\right)^2 - \frac{4}{LC}}}{2C}$

$$= \frac{-1}{2RC} \pm \left[\sqrt{\left(\frac{1}{2R}\right)^2 - \frac{1}{LC}} \right]$$

$$V_n(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Let $\alpha = \frac{1}{2RC}$, $\omega_0 = \frac{1}{LC}$

CASES:-

- ① $\alpha^2 > \omega^2$ Over damped case (for real solution)
 for this case we have to find constants A_1 & A_2 .
 These are found by using initial conditions $V(0)$ & $I(0)$. Another eqⁿ is developed by putting $t=0$ in when the differential equation is not of 2nd order.

$I(0)$ & $V(0)$ is given. $\therefore A_1$ & A_2 is to be found
 1st $V(0) = A_1 + A_2$ by ~~diff~~ $V_n(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

2nd $\frac{V_0}{R} + i(0) + \left(\frac{dV_0}{dt} \right) = 0$

find this using $\frac{dV_n}{dt}(0) = A_1 s_1 + A_2 s_2$
 $\therefore$ 2nd eqⁿ will also give eqⁿ of A_1 & A_2

② $s_1 = s_2 = -\alpha$ $\left(\alpha = \frac{1}{2RC} \rightarrow \omega_0^2 = \frac{1}{LC} \right)$

$V_n(t) = (A_1 t + A_2) e^{-\alpha t}$

$$\boxed{\omega_0^2 > \alpha^2}$$

$$\boxed{\omega_d = \sqrt{\omega_0^2 - \alpha^2}}$$

③. $x_n(t) = A_1 e^{st} + A_2 e^{s^* t}$

$$x_n(t) = e^{-\alpha t} \left[A_1 \cos \omega_d t + j A_1 \sin \omega_d t + A_2 \cos \omega_d t - j A_2 \sin \omega_d t \right]$$

$$= e^{-\alpha t} \left[\underbrace{(A_1 + A_2)}_{B_1} \cos \omega_d t + j \underbrace{(A_1 - A_2)}_{B_2} \sin \omega_d t \right]$$

$$\rightarrow x_n(t) = e^{-\alpha t} [B_1 \cos \omega_d t + B_2 \sin \omega_d t]$$

④. $\boxed{v_n(0) = x_n(0) = B_1}$

(i) find $\frac{dx_n(t)}{dt} \big|_{t=0}$.

$$\frac{dx_n(t)}{dt} = e^{-\alpha t} \left[(\omega_d B_2 - \alpha B_1) \cos \omega_d t - (\omega_d B_1 + \alpha B_2) \sin \omega_d t \right]$$

at $t=0$.

$$\rightarrow \frac{dx_n(0)}{dt} = \omega_d B_2 - \alpha B_1$$